

Akshitha Reddy

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EDUCATION

- Marri Laxman Reddy Institute of Tech & Management** Hyderabad, Telangana
Bachelor of Technology (BTech); CGPA: 8.2 2019 – 2023

SKILLS

Programming Languages: SQL, Python

Machine Learning & Deep Learning: Supervised/Unsupervised Learning, Regression (Linear, Logistic), Tree-based Models (Random Forest, XGBoost, LightGBM), ANN, Clustering (KMeans++, DBSCAN, GMM), Dimensionality Reduction (PCA, t-SNE), Time Series Analysis

Generative AI: RAG Applications, Embeddings, Vector Databases (Chroma DB), Prompt Engineering, LLM Fine-tuning

Statistics & Data Analysis: Descriptive/Inferential Stats, Hypothesis Testing, A/B Testing, Gain/Lift Charts, KS Scores

Libraries & Frameworks: Pandas, NumPy, Scikit-Learn, SciPy, Matplotlib, TensorFlow

Tools & Platforms: Jupyter Notebook, Git, MS Excel, Power BI

EXPERIENCE

- ADP** Hyderabad, Telangana
Data Scientist Sep 2024 – Current
 - Developed attrition prediction models using regression & tree-based classifiers to quantify turnover risk, achieving 18% reduction in attrition and \$2.1M annual savings in rehiring costs.
 - Built employee career progression models leveraging promotion velocity, tenure patterns, and compensation parity to forecast career growth trajectories and internal mobility opportunities.
 - Extracted and transformed large HR and Payroll datasets from enterprise data sources using SQL and ETL pipelines, enabling data-driven insights and improving reporting efficiency.
 - Performed advanced SQL querying, data cleansing, validation and mapping on workforce data, ensuring high quality and supporting analytics-ready datasets for business decision-making.
- Hanyaa Auto Technologies** Hyderabad, India
Data Scientist Oct 2023 - Sep 2024
 - Developed and deployed machine learning models to **predict song success (Hit/Flop)** using historical audio features, metadata, and user engagement data, improving predictive accuracy through feature engineering and model optimization.
 - Designed and implemented an **audio plagiarism detection system** using audio signal processing and machine learning techniques to identify similarities between songs and detect potential copyright infringements.
 - Performed end-to-end data processing, including data cleaning, preprocessing, exploratory data analysis (EDA), feature extraction, and feature engineering on large-scale audio datasets.
 - Extracted and engineered advanced audio features, including **MFCCs, spectral features, chroma vectors, tempo, rhythm descriptors, and acoustic characteristics** to enhance model performance.
 - Built and evaluated supervised machine learning models for classification tasks using algorithms such as **Random Forest, XGBoost, Support Vector Machines (SVM), and Logistic Regression**.
 - Conducted hyperparameter tuning, cross-validation, and model performance analysis using metrics such as **Accuracy, Precision, Recall, F1-Score, and ROC-AUC** to improve robustness and generalization.
 - Utilized Python and data science libraries including **Pandas, NumPy, Scikit-learn, Librosa, Matplotlib, and Seaborn** for data analysis, feature engineering, model development, and visualization.
 - Documented model development processes, experimental results, and technical findings to support reproducibility and knowledge sharing.
- Deloitte** Hyderabad, India
Apprenticeship Jul 2023 - September 2023
 - Supported documentation, reporting, and general administrative tasks.
 - Gained exposure to corporate workflows and professional project environments.

PROJECTS

- **Song Success Prediction Using Audio Analytics**

- ***Classification — Audio Signal Processing — Feature Engineering — XGBoost — Model Evaluation — EDA***

- Developed a **machine learning classification model** to predict whether a new song would be a **Hit or Flop** using historical audio features and song metadata.
- Extracted and engineered **audio features** including MFCCs, chroma, spectral features, tempo, and rhythm descriptors using **Librosa** and Python.
- Performed **EDA, feature engineering, hyperparameter tuning, and cross-validation** to optimize model performance and improve prediction accuracy.
- Built and evaluated models including **XGBoost, Random Forest, and SVM**, leveraging feature importance analysis to identify key drivers of song success.

- **Song Plagiarism Detection Using Audio Analytics**

- ***Audio Signal Processing - Similarity Detection - Feature Engineering - Machine Learning***

- Built an **audio plagiarism detection system** to identify copied tunes and measure similarity between songs using audio signal processing and machine learning techniques.
- Extracted features such as **MFCCs, chroma, spectral, tempo, and rhythm descriptors** to capture musical characteristics and improve similarity detection.
- Developed a **similarity scoring framework** that detected potential plagiarism and quantified song-to-song resemblance through percentage-based similarity scores.
- Evaluated and optimized detection performance using feature engineering and similarity analysis on large-scale music datasets.